Q9. Psychological Creativity through Evolutionary Algorithms, and the Step to Historical Creativity

> **Question.** Discuss different possibilities where psychological (P) creativity is achieved through an evolutionary algorithm. Can it be extended to historical (H) creativity? What could the challenges be? Think of a few ideas to resolve them. How can we expand the conceptual space in this case?

1. The idea in plain words

An evolutionary algorithm (EA) keeps making new designs by mixing and mutating old ones. Almost everything it makes is **new to itself**. That is P-creativity (Boden): new to the individual or agent. H-creativity is much harder: the idea must be **new to all of history** and still useful. The EA cannot see history unless we show it. So the honest answer is: yes, it can be extended, but only if we (1) give the EA a memory of what the world has already made and (2) let it step out of the old conceptual space.

2. How we measure the two kinds of creativity

Kind	Test used in this answer
P-creative	Distance to the nearest design in the agent's **own archive** > 0.12, so the agent has never made anything like it
H-creative	Distance to the nearest of **300 historical designs** > 0.40 **and** usefulness > 0.70. It is new to the world *and* valuable

Every H-creative design is also P-creative, but not the other way round. In other words, P-creativity is H-creativity judged against a much smaller archive.

3. Possibilities where P-creativity comes from evolution

Area	What evolves	Why it is P-creative
Vehicle / product design	Body and wheel genes (the car GA from the lectures)	Crossover gives body/wheel mixes the agent never drew before
Poetry	Word sequences (POEVOLVE, McGonnagall)	New lines that fit rhyme and metre
Music	Jazz riffs (Al Biles' GenJam)	Each solo differs from the bot's earlier ones
Chatbots and rules	Decision trees (see Q1)	New dialogue strategies for the bot
Neural networks	Connection matrices, learning rules (Chalmers 1990)	New architectures for that system
Art and patterns	Drawing programs (see Q8)	New shapes from combined actions

4. Assumptions for the experiment

- A design is a small vehicle with 6 genes in [0, 1]: length, height, wheel size, wheel gap, roof slope and colour.
- **Usefulness** checks simple engineering rules: wheels near the ends, wheels sized to the body, roof not too steep.

- **History** = 300 known designs around 5 prototypes (sedan, SUV, truck, bus, sports car).
- EA: 40 designs per generation, 40 generations, tournament selection, blend crossover, Gaussian mutation, 2 elites.
- Three runs, all with the same seed:
  - **P-only**: fitness = usefulness + novelty against the agent's own archive.
  - **global**: from generation 21, fitness also rewards distance to history (a global archive).
  - **global+expand**: as "global", plus a 7th gene from generation 21, a "smart IoT skin" (sensor coverage) that no historical design has.

5. Step-by-step approach

1. Load the history archive H and start with an empty own archive P.
2. Create a random population and score each design: fitness = usefulness + novelty.
3. Label each design: P-creative if far from P (then store it in P), H-creative if also far from H and useful.
4. If the H-creative rate stays low, apply the fixes: reward distance to H, and expand the conceptual space with a new gene.
5. Select, cross over and mutate to make the next generation. Repeat until the last generation.

Evolutionary loop with P- and H-creativity checks

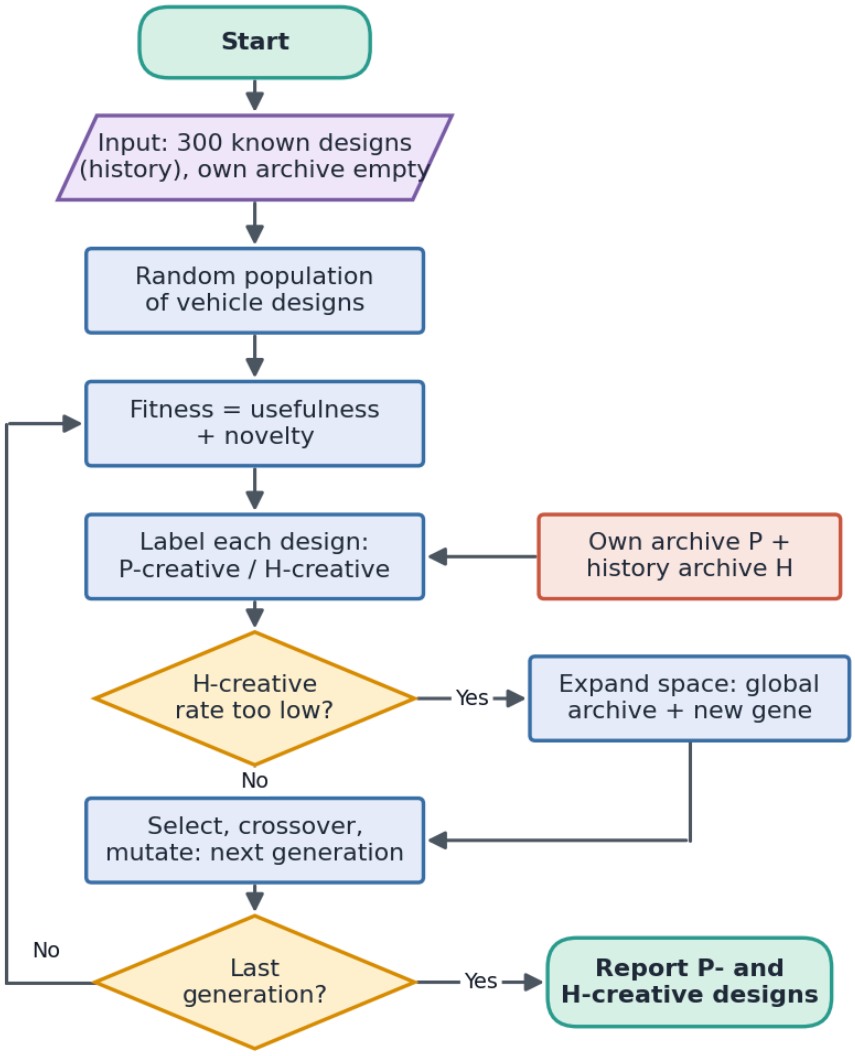


Figure 1. Flowchart of the EA with P- and H-creativity checks.

6. Pseudocode

```
function CreativeEA(history H, generations G, switch S):
    P ← { };  population ← random designs
    for g = 1 to G:
        if g = S + 1: add a new gene to every design            # expand the conceptual space
        for each design x in population:
            if dist(x, P) > τ_P:
                P.add(x)                                        # P-creative
                if dist(x, H) > τ_H and useful(x) > 0.7: mark x H-creative
        fitness ← useful(x) + 0.5·novelty(x, P)
        if g > S: fitness ← useful(x) + 0.3·novelty(x, P) + 0.6·novelty(x, H)   # global archive
        population ← elites + mutate(blend_crossover(tournament(population)))
    return P-creative and H-creative designs
```

7. Worked example (real run)

Results of the "global+expand" run, in blocks of 5 generations:

Generations	P-creative	H-creative	H share	What is happening
1–5	130	11	8.5%	Everything is new to a "newborn" agent
6–10	46	3	6.5%	Own archive fills up
11–15	21	0	0.0%	Useful designs drift back toward history
16–20	15	1	6.7%	Stuck: P-creative but not H-creative
21–25	60	12	20.0%	Fixes switched on
26–30	60	26	43.3%	The new gene is being explored
31–35	**47**	**37**	**78.7%**	**Most new designs are now new to history**
36–40	42	30	71.4%	Stable

Totals after 40 generations for the three runs:

Run	P-creative	H-creative	H share
P-only (plain EA)	276	18	6.5%
+ global archive	294	33	11.2%
+ global archive + new gene	**421**	**120**	**28.5%**

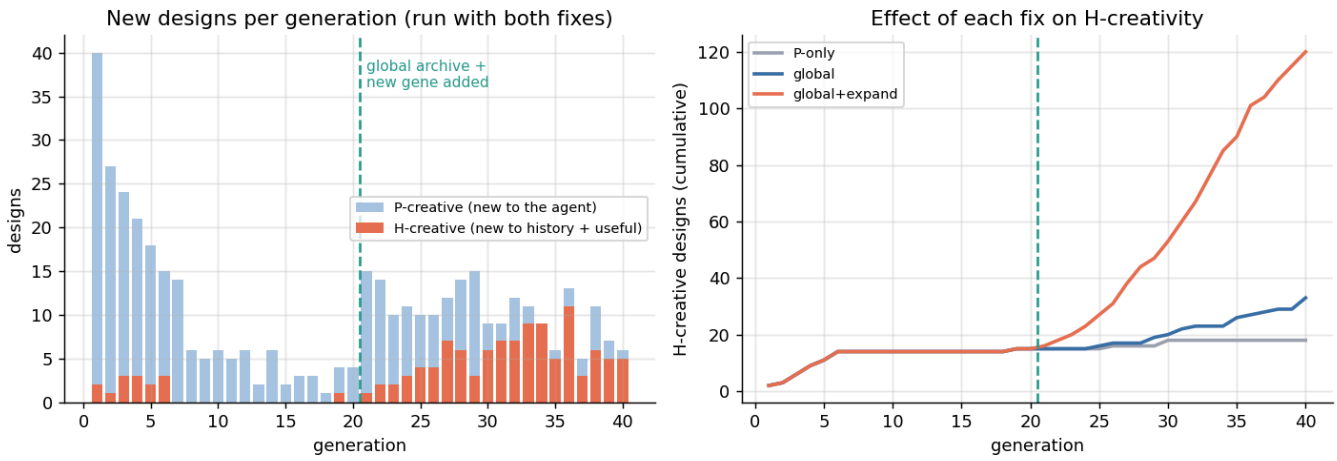


Figure 2. Left: P-creative (blue) vs H-creative (orange) designs per generation. Right: cumulative H-creative designs for the three runs.

8. Can it be extended to historical creativity?

Yes, but not for free. The plain EA made 276 P-creative designs, yet only 18 (6.5%) were new to history and useful. The reason is simple: the useful region of the old space is already crowded with sedans, SUVs and trucks, so evolution keeps "rediscovering" them. Once the EA could see history (global archive) and had a new dimension to explore, it produced **120 H-creative designs, 6.7 times more**. Most of them sit where no known vehicle sits: short but tall bodies (Figure 3, right).

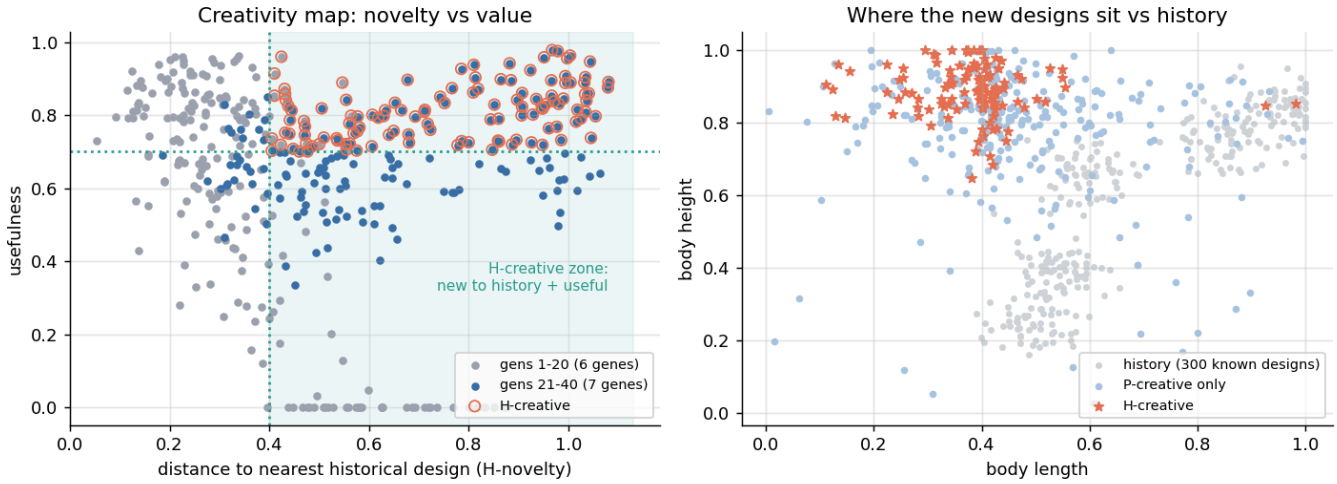


Figure 3. Left: novelty vs usefulness; the shaded corner is the H-creative zone. Right: new designs vs the 300 historical designs (grey).

9. Challenges

- **Knowing history.** No archive holds every design ever made, so an "H-creative" label is only as good as the archive.
- **Judging value.** Usefulness for brand-new ideas is uncertain. History often decides value later (a social judgement).
- **Novelty vs usefulness.** Random designs are novel but useless (the grey points at usefulness 0 in Figure 3).
- **Pull back to the known.** Fitness rules are written by people, so they favour what already exists (generations 11–20).
- **Premature convergence** and homogenisation. AI can raise individual creativity yet reduce collective diversity.

- **Cost.** Comparing every design with a large history archive and validating with experts is slow.

10. Ideas to resolve them

Idea	How it helps
Global archive + novelty search	Reward distance to *history*, not just to the agent's memory (+15 H-creative in our run)
Multi-objective fitness	Keep novelty, usefulness and surprise as separate goals (a Pareto front) so neither hides the other
Island models	Several sub-populations evolve apart and swap genes now and then, which keeps diversity
Human-in-the-loop (creativity test)	Experts rate candidates in the H-creative zone. The machine proposes; people judge value
Embedding search over patents/papers	A practical way to build a big history archive
Transformational operators	Change the representation itself, not only gene values (next section)

11. How can we expand the conceptual space?

- **Add a new dimension (gene).** In our run, a "smart IoT skin" gene appeared at generation 21. No historical vehicle has it, so useful designs that use it are new to history. The H share jumped from about 7% to **79%**. This is the course idea: *adding a sensor modality changes the space, not merely the data*.
- **Release boundaries of compatibility.** Relax rules such as "exactly two axles", like the pink-elephant overrule.
- **Change the rules R and traversal T** (Wiggins). Allow new operators such as nesting or analogy from other domains.
- **Combine spaces from other domains** (systemic view): vehicle + drone, or vehicle + furniture.
- **Generalise properties** as in the "GENERATE" slide. "Some shape on two circles" invites a pentagon on wheels.

12. Code and real run

`q09_code.py` runs all three EAs, counts P- and H-creative designs and draws the figures.

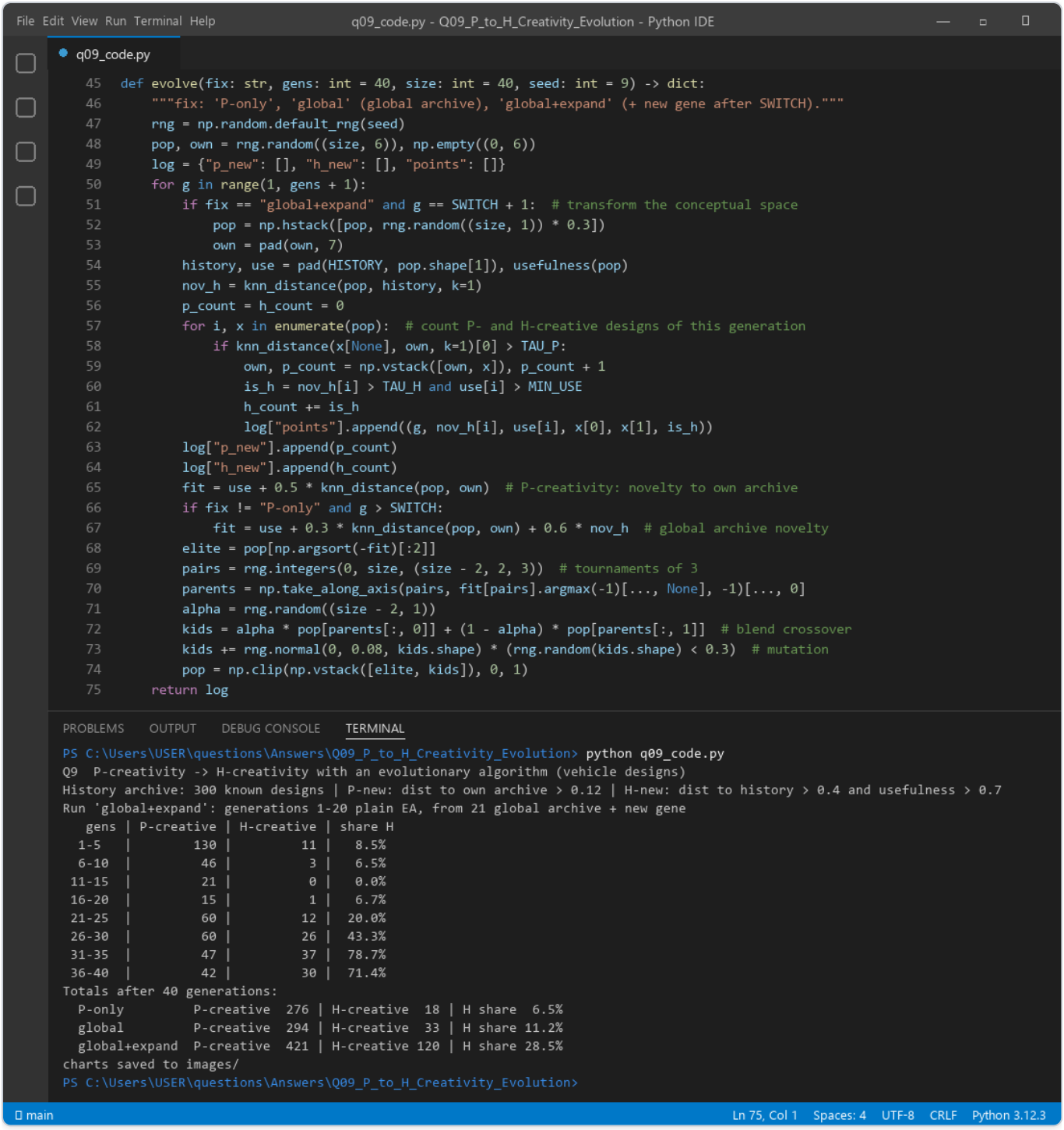


Figure 4. The `evolve()` function and the real terminal output.

13. Summary

> Evolution gives P-creativity easily: 276 designs new to the agent. H-creativity needs more, because only 6.5% of those were new to history and useful. Giving the EA a global archive and expanding its conceptual space with a new dimension raised that to 120 H-creative designs. Final value judgement should still involve people.

Files in this folder: `q09_code.py` (source), `q09_code.ipynb` (Colab/Jupyter version, already run), `q09_output.txt` (console output), `images/` (all figures).